

A proof of Kalai’s tight-tree conjecture

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Abstract

We prove Kalai’s conjecture for tight uniform hypergraph trees. If T is a tight r -uniform tree with m edges, where $r \geq 2$, then every T -free r -uniform hypergraph H satisfies $re(H) \leq (m-1)|\partial H|$, where ∂H is the family of $(r-1)$ -sets contained in its edges. The proof extends the permutation-counting argument of the recent Erdős–Sós proof obtained in the FrontierMath Erdős benchmark. We also give a direct proof for tight paths and a Lean 4 formalization of the general theorem.

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ChatGPT 6 Astra was used to develop the argument, write the exposition, and produce the accompanying Lean formalization. The author’s role was minimal, consisting mostly of asking questions towards trying to understand the recent Erdős–Sós proof and its possible extensions.

1 Introduction

In 1984, Gil Kalai [1] proposed a hypergraph extension of the Erdős–Sós conjecture. He conjectured that every n -vertex r -uniform hypergraph with more than $\frac{m-1}{r} \binom{n}{r-1}$ edges contains every tight r -uniform tree with m edges. The conjecture was published by Frankl and Füredi [2, Conjecture 3.6], who attribute it to a 1984 communication from Kalai. We prove the conjecture, in the sharper form that replaces $\binom{n}{r-1}$ by the number of $(r-1)$ -sets contained in edges of the hypergraph.

Our proof is based on the new proof of the Erdős–Sós conjecture obtained in the FrontierMath Erdős benchmark and recorded in [7].¹ We adapt that proof’s counting of marked vertex permutations and induction on the target tree; we do not claim an independent origin for this method. In the graph argument, copies are rooted at a specified vertex. For the hypergraph argument, we prescribe the images of all vertices in one edge. This allows copies of branches attached through different faces to agree on their common edge. The counting inequalities and the required decomposition are proved here for hypergraphs, rather than deduced by applying the graph theorem as a black box.

1.1 Definitions and main result

All hypergraphs are finite and simple. An r -uniform hypergraph, or r -*graph*, has edges that are r -element vertex sets. A copy is injective and edge-preserving, but need not be induced.

¹The repository is maintained by Tom Adamczewski. Its provenance account credits the graph proof to GPT-6 Astra, working autonomously in the FrontierMath Erdős benchmark of Adamczewski and Bloom.

Definition 1.1. A *tight r -tree* with $m \geq 1$ edges admits an edge ordering $e_1, \dots, e_m$ such that, for every $j \geq 2$, there is a vertex $x_j \in e_j \setminus \bigcup_{h < j} e_h$ and an index $p(j) < j$ with $e_j \setminus \{x_j\} \subseteq e_{p(j)}$. The vertex set of a tight tree is the union of its edges; in particular, there are no additional isolated vertices. Such a tree has $m + r - 1$ vertices. The *tight path* $P_m^{(r)}$ has distinct vertices $x_1, \dots, x_{m+r-1}$ and edges

$$\{x_j, x_{j+1}, \dots, x_{j+r-1}\}, \quad 1 \leq j \leq m.$$

Throughout the paper, m denotes the number of *edges* of the target. We write $e(H) = |E(H)|$. The $(r - 1)$ -*shadow* of an r -graph H is

$$\partial H = \{F : |F| = r - 1, F \subseteq e \text{ for some } e \in E(H)\}.$$

Our main result is the following.

Theorem 1.2 (Tight-tree shadow bound). *Let $r \geq 2$ and $m \geq 1$, and let T be a tight r -tree with m edges. Every T -free r -graph H satisfies*

$$re(H) \leq (m - 1)|\partial H|. \quad (1.1)$$

Since $|\partial H| \leq \binom{|V(H)|}{r-1}$, this proves Kalai's conjecture in its usual form; see [3, 6].

Corollary 1.3 (Kalai's bound). *Let $r \geq 2$ and $m \geq 1$, and let T be a tight r -tree with m edges. Every n -vertex T -free r -graph H satisfies*

$$e(H) \leq \frac{m - 1}{r} \binom{n}{r - 1}.$$

The shadow formulation appears as Conjecture 2.1 in [3]. Its coefficient is sharp: for $m \geq 2$, the complete r -graph on $m + r - 2$ vertices has no copy of T and attains equality in (1.1). The case $r = 2$ gives the corresponding tree bound with the number of nonisolated host vertices in place of $|V(H)|$.

1.2 Related work

Frankl and Füredi [2, Theorem 3.8] proved Kalai's conjecture for star-shaped tight trees: those containing an edge that meets every other edge in $r - 1$ vertices. Füredi, Jiang, Kostochka, Mubayi, and Verstraëte [3] proved Kalai's bound for tight trees with at most four edges and an asymptotic form for tight trees of bounded trunk size. Their work also develops bounds in terms of the shadow of the host. For tight paths, their geometric method in [4] gives improved upper bounds in arbitrary uniformity. In particular, [4, Theorem 1.2] gives

$$\text{ex}(n, P_m^{(3)}) \leq \frac{1}{2} \left(m + \left\lfloor \frac{m - 1}{3} \right\rfloor \right) \binom{n}{2}.$$

Here $\text{ex}(n, F)$ denotes the maximum number of edges in an n -vertex F -free hypergraph of the same uniformity as F . Stein [6] proved Kalai's conjecture for r -partite r -graphs. Bright, Milans, and Porter [5] study the related problem of excluding tight paths under order-preserving embeddings. No partiteness or order-preservation restriction is imposed here.

1.3 Outline of the proof

Consider a permutation $(v_1, \dots, v_n)$ of the host vertices. We mark position $i \geq r$ if v_i together with the first $r - 1$ vertices forms an edge. We count pairs consisting of a permutation and one of its marked positions. For a fixed tree, we count those pairs for which the first i vertices contain a copy of the tree, with one specified edge mapped to the marking edge. The copy need not follow the order of the permutation, and each pair is counted only once, even if it admits several copies.

The induction compares these counts for smaller trees. To add a leaf, we use a copy lying entirely before the current marked position; the vertex at that position is then available for the new leaf. This excludes at most one pair for each permutation. To join two trees along a common edge, we require their copies to agree on that edge and use disjoint sets of vertices elsewhere. An injection obtained by reordering the permutation gives the needed counting inequality. Finally, we show that every tight tree can be decomposed using these two operations. The resulting inequality gives the claimed bound when the host contains no copy of the tree.

Section 2 gives the marked-position argument for 3-uniform paths, and Section 3 treats arbitrary uniformity. Sections 4 and 5 count copies with a specified edge and prove the two counting inequalities. Section 6 completes the induction. Section 7 records an example and sharpness, and Section 8 describes the accompanying formal proof.

2 Marked positions for tight paths

We first prove the theorem for 3-uniform paths. In this case, the injection rearranges just three entries of a permutation. The main point is to use a path that avoids the vertex at the current marked position, so that this vertex can extend the path.

2.1 Marked permutations

Fix a 3-graph H on $n \geq 3$ vertices. For distinct vertices a, b , write

$$d_H(ab) = |\{z : \{a, b, z\} \in E(H)\}|.$$

For a permutation $\pi = (v_1, \dots, v_n)$ and $3 \leq i \leq n$, call (π, i) *marked* if $\{v_1, v_2, v_i\} \in E(H)$. Its prefix is the vertex set $S_i(\pi) = \{v_1, \dots, v_i\}$. Let $\mathcal{O}(H)$ be the set of marked states and $M = |\mathcal{O}(H)|$.

For $k \geq 1$, let $\mathcal{A}_k$ be the set of marked states whose prefixes contain a tight k -edge path with vertex sequence

$$v_1, v_2, y_3, \dots, y_{k+2}. \tag{2.1}$$

The starting pair is ordered, but the remaining path vertices need not occur in the permutation in path order. Support is an existence condition inside the prefix vertex set. Each supporting state is counted once, independently of the number of witnesses. Write $A_k = |\mathcal{A}_k|$.

Lemma 2.1 (Basic counts). *With the notation above,*

$$A_1 = M = 6e(H)(n-2)!. \tag{2.2}$$

Let $\mathcal{Q}$ be the set of permutations whose first pair belongs to ∂H . Then

$$Q := |\mathcal{Q}| = 2|\partial H|(n-2)!. \tag{2.3}$$

Proof. Every marked state supports a one-edge path starting at its first pair, so $\mathcal{A}_1 = \mathcal{O}(H)$. For each fixed ordered pair (a, b) , there are $(n-2)!$ permutations starting with (a, b) , and each contributes $d_H(ab)$ marks. Each edge contributes to six ordered pairs, giving

$$M = (n-2)! \sum_{a \neq b} d_H(ab) = 6e(H)(n-2)!.$$

Each shadow pair has two orders and each such order can begin $(n-2)!$ permutations. This proves (2.3). $\square$

2.2 Extending a path

If an earlier marked position already admits the required path, that path avoids the vertex at the current marked position. We therefore discard the first such position in each permutation and extend the paths at all remaining positions. This gives the following inequality.

Lemma 2.2 (Path extension). *For every $k \geq 1$,*

$$A_k \leq A_{k+1} + Q. \quad (2.4)$$

Proof. For each permutation π contributing to $\mathcal{A}_k$, let $i_0(\pi)$ be its first position in $\mathcal{A}_k$, and put

$$\mathcal{D}_k = \{(\pi, i_0(\pi)) : \pi \text{ contributes to } \mathcal{A}_k\}.$$

Such a permutation has a marked position, so its first pair lies in ∂H . Hence $|\mathcal{D}_k| \leq Q$.

We construct an injection $\Phi_k : \mathcal{A}_k \setminus \mathcal{D}_k \rightarrow \mathcal{A}_{k+1}$. Take a state (π, i) in its domain and write

$$\pi = (a, b, W, z, R), \quad W = (v_3, \dots, v_{i-1}), \quad z = v_i, \quad R = (v_{i+1}, \dots, v_n).$$

Since $i > i_0(\pi)$, the prefix $S_{i_0(\pi)}(\pi)$ contains a path

$$a, b, y_3, \dots, y_{k+2}$$

and does not contain z . The marking triple $\{z, a, b\}$ extends this to a path

$$z, a, b, y_3, \dots, y_{k+2} \quad (2.5)$$

with $k+1$ edges. Indeed, its first edge is $\{z, a, b\}$ and its remaining consecutive triples are precisely the old path edges. All vertices remain distinct.

Define

$$\Phi_k((a, b, W, z, R), i) = ((z, a, W, b, R), i). \quad (2.6)$$

The new prefix has the same vertex set. Its first pair is (z, a) , the starting pair of (2.5), and its last vertex is b . Thus its marking triple is still $\{z, a, b\}$. The image is therefore in $\mathcal{A}_{k+1}$.

The retained index i identifies the displaced entry b . For any image (ρ, i) , the unique possible source ordering is

$$(\rho_2, \rho_i, \rho_3, \dots, \rho_{i-1}, \rho_1, \rho_{i+1}, \dots, \rho_n). \quad (2.7)$$

This inverse cycle on positions $1, 2, i$ recovers the source without using the earlier index $i_0(\pi)$ or choosing a path witness. Hence $A_k - |\mathcal{D}_k| \leq A_{k+1}$ proves (2.4). $\square$

Theorem 2.3 (Tight-path shadow bound). *For every integer $m \geq 1$ and every $P_m^{(3)}$ -free 3-graph H ,*

$$3e(H) \leq (m-1)|\partial H|. \quad (2.8)$$

In particular, for an n -vertex host, $e(H) \leq \frac{m-1}{3} \binom{n}{2}$.

Proof. If $n < 3$ there are no edges, so assume $n \geq 3$. Iterating Lemma 2.2 gives

$$6e(H)(n-2)! = A_1 \leq A_m + (m-1)Q.$$

The forbidden-path hypothesis gives $A_m = 0$; substitution of (2.3) and division by $2(n-2)!$ gives (2.8). For $m = 1$ the iteration is empty and $A_1 = 0$, which gives the same conclusion. Finally, $|\partial H| \leq \binom{n}{2}$. $\square$

2.3 The role of the mark

The path and the marking edge need not start with the same edge. The path only has to lie among the first i vertices and start with the ordered pair (v_1, v_2) . The mark supplies another edge through that pair. Moving z to the first position and b to the marked position preserves both requirements. The map does not depend on which path is chosen; the earlier marked position ensures that a path avoiding z exists.

Example 2.4. For five distinct vertices a, b, c, d, z , write abc for $\{a, b, c\}$, and similarly for other edges. Suppose H contains abc, bcd, zab , and take

$$\pi = (a, b, c, d, z), \quad i = 5.$$

The prefix at position 4 contains the two-edge path a, b, c, d . However, position 4 is marked only if $abd \in E(H)$. Without that edge, position 5 is the first supporting mark and is discarded. If $abd \in E(H)$, the map is defined at position 5 and gives

$$(a, b, c, d, z) \mapsto (z, a, c, d, b),$$

whose full prefix supports z, a, b, c, d and whose mark is zab . Thus an earlier supporting prefix alone is not sufficient: the proof uses an earlier supporting *marked state*.

Remark 2.5 (Boundary cases and equality). An empty host has $A_k = M = Q = 0$. For $n = 3$, every permutation has at most one mark, so every supporting pair is discarded. For the complete 3-graph on $s = m + 1 \geq 3$ vertices, there is no $P_m^{(3)}$, and

$$3 \binom{s}{3} = (m-1) \binom{s}{2}.$$

Thus the coefficient in the shadow bound cannot be reduced. More precisely, every permutation has $s - 2$ marks and exactly $s - k - 1$ of them support a k -edge path, for $1 \leq k \leq s - 1$. Hence $A_k = (s - k - 1)s!$ and $A_k - A_{k+1} = Q = s!$ throughout the nontrivial iteration. Adding isolated vertices does not change the shadow or edge count and only changes the factorial normalizations.

3 Tight paths in arbitrary uniformity

The preceding proof extends by replacing the three-cycle with an r -cycle of entries. We give the map explicitly. Fix $r \geq 2$ and $n \geq r$. Mark (π, i) , $i \geq r$, when

$$\{v_1, \dots, v_{r-1}, v_i\} \in E(H).$$

Let $A_k^{(r)}$ count the marked states whose prefixes support a tight k -edge path starting with the ordered tuple $(v_1, \dots, v_{r-1})$. The same normalization gives

$$A_1^{(r)} = r! e(H)(n - r + 1)!, \quad Q_r = (r - 1)! |\partial H| (n - r + 1)!. \quad (3.1)$$

Here Q_r counts permutations whose first $(r - 1)$ -set belongs to the shadow.

Discard the first supporting marked state of each permutation. At a remaining state, write

$$\pi = (a_1, \dots, a_{r-1}, W, z, R).$$

An earlier path beginning $a_1, \dots, a_{r-1}$ avoids z , and prepending z adds exactly the marking edge. Apply

$$((a_1, \dots, a_{r-1}, W, z, R), i) \mapsto ((z, a_1, \dots, a_{r-2}, W, a_{r-1}, R), i). \quad (3.2)$$

The prefix set and marking r -set are unchanged. The new first $(r - 1)$ -tuple is the starting tuple of the extended path. The inverse takes the entries at positions $2, \dots, r - 1, i$ as the old root tuple, restores the entry at position 1 to position i , and leaves all other entries fixed. Thus

$$A_k^{(r)} \leq A_{k+1}^{(r)} + Q_r.$$

When $r = 2$, the tuple $(a_1, \dots, a_{r-2})$ is empty and the map simply exchanges the first entry and the mark.

Corollary 3.1. *If an r -graph H contains no tight path with m edges, then*

$$re(H) \leq (m - 1) |\partial H|.$$

Consequently it satisfies Corollary 1.3 for the target $P_m^{(r)}$.

Proof. For $n < r$ there are no edges. Otherwise iterate the preceding inequality, use $A_m^{(r)} = 0$, and divide (3.1) by $(r - 1)!(n - r + 1)!$. $\square$

4 Copies with a specified edge

For paths, it was enough to specify the first $r - 1$ vertices of a copy. A tight tree may have several edges attached through different $(r - 1)$ -subsets of the same edge. For example, the tight 3-tree

$$123, \quad 124, \quad 135, \quad 236$$

has edges attached through all three pairs of 123. To combine copies of these branches, we must make them agree on the images of all three vertices of 123, not just two of them.

We therefore choose an edge of the target, called its *root edge*, and specify the image of each of its vertices. Unlike the condition used for paths, we now require this chosen edge to map to the marking edge. Copies of two branches can then be made to agree on their common edge. In the proof of Lemma 5.2, we arrange for their other vertices to lie in disjoint parts of the permutation.

4.1 Definitions and basic counts

Fix $r \geq 2$ and an r -graph H on $n \geq r$ vertices. Let

$$\mathcal{O}_r(H) = \{(\pi, i) : \pi = (v_1, \dots, v_n), r \leq i \leq n, \{v_1, \dots, v_{r-1}, v_i\} \in E(H)\}.$$

Put $S_i(\pi) = \{v_1, \dots, v_i\}$ and let $\mathcal{Q}_r$ be the set of permutations whose first $(r-1)$ -set lies in ∂H . Throughout the general-tree argument write

$$M = |\mathcal{O}_r(H)| = r! e(H)(n-r+1)!, \quad Q = |\mathcal{Q}_r| = (r-1)! |\partial H|(n-r+1)!. \quad (4.1)$$

To verify (4.1), each ordered $(r-1)$ -tuple begins $(n-r+1)!$ permutations, each with one mark for every edge containing that tuple. Summing over the ordered tuples gives $r!e(H)$. Counting the orders of each shadow face gives the formula for Q .

Definition 4.1 (Support with a specified edge). Let T be an r -graph with a distinguished ordered edge $\mathbf{e} = (u_1, \dots, u_r)$, whose underlying set is e . A marked state (π, i) *supports* $(T, \mathbf{e})$ if there is an injective edge-preserving map $\varphi : V(T) \rightarrow S_i(\pi)$ with

$$\varphi(u_j) = v_j \quad (1 \leq j < r), \quad \varphi(u_r) = v_i. \quad (4.2)$$

Let $\mathcal{A}_H(T, \mathbf{e})$ be the set of these states and put $R_H(T, \mathbf{e}) = |\mathcal{A}_H(T, \mathbf{e})|$. Each state is counted once, regardless of the number of witnessing embeddings. We omit the host subscript when it is fixed.

Lemma 4.2 (Independence of the ordering of the root edge). *The count $R(T, \mathbf{e})$ depends on the root edge e , but not on the ordering of its vertices. We may therefore denote it by $R(T, e)$.*

Proof. Fix the marked position i . If the target root order is changed by a permutation of its r entries, apply the same permutation to the host entries in positions $1, \dots, r-1, i$. All other entries and the index i remain fixed. The prefix vertex set and the marking edge are unchanged. The same embedding sends each root vertex to its required position after the change. The inverse permutation reverses the construction, so this is a bijection for each i , and hence for their disjoint union. $\square$

Thus reordering the root vertices does not change the total count. It can change whether an individual pair (π, i) supports the target, so we will specify the root orders when comparing sets of pairs.

5 Extending and joining copies

We need two inequalities: one for adding a leaf edge, and one for joining two hypergraphs along an edge. The second proof is easier to write after moving the marking edge to the first r positions. We first show that this rearrangement leaves the counts unchanged.

5.1 A new leaf edge and a change of root

Lemma 5.1 (Adding a leaf edge). *Let S have a distinguished edge $f = F \cup \{p\}$, where $|F| = r-1$. Let $z \notin V(S)$, and form T by adding z and the edge $e = F \cup \{z\}$ to S . Then*

$$R(S, f) \leq R(T, e) + Q. \quad (5.1)$$

Thus the distinguished edge changes from f to the newly added edge e .

Proof. Order F as $(u_1, \dots, u_{r-1})$ and use the orders $\mathbf{f} = (u_1, \dots, u_{r-1}, p)$ and $\mathbf{e} = (u_1, \dots, u_{r-1}, z)$. Lemma 4.2 allows these choices without changing either count.

For each permutation π having a state in $\mathcal{A}(S, \mathbf{f})$, discard its first such state, at position $i_0(\pi)$. Let $\mathcal{D}$ be the set of discarded states. Each contributing permutation has a mark, so belongs to $\mathcal{Q}_r$; hence $|\mathcal{D}| \leq Q$.

Take $(\pi, i) \in \mathcal{A}(S, \mathbf{f}) \setminus \mathcal{D}$. An earlier state (π, i_0) supports S , with $u_j \mapsto v_j$ for $j < r$ and $p \mapsto v_{i_0}$. This earlier embedding avoids v_i because $i_0 < i$. Extend it by setting $z \mapsto v_i$. The only added target edge maps to the current marking edge $\{v_1, \dots, v_{r-1}, v_i\}$. The extended embedding is injective and sends each vertex of e to its required position.

Consequently the *identity map on states* gives the inclusion

$$\mathcal{A}(S, \mathbf{f}) \setminus \mathcal{D} \subseteq \mathcal{A}(T, \mathbf{e}).$$

Taking cardinalities proves (5.1). The current state's witness for S need not be used; it is the earlier witness that guarantees avoidance of the new vertex. $\square$

5.2 Moving the marking edge to the front

Consider the following alternative set of pairs:

$$\mathcal{B}_r(H) = \{(\rho, i) : \rho = (w_1, \dots, w_n), r \leq i \leq n, \{w_1, \dots, w_r\} \in E(H)\}.$$

The first r entries form an ordered edge, and i specifies how many vertices may be used in a copy. Every value $r, \dots, n$ is allowed. We indicate the end of the prefix by a vertical bar after position i , which we call the *divider*. The bijection between the two sets of pairs is

$$\begin{aligned} \Theta : \mathcal{O}_r(H) &\longrightarrow \mathcal{B}_r(H), \\ ((a_1, \dots, a_{r-1}, W, c, C), i) &\longmapsto ((a_1, \dots, a_{r-1}, c, W, C), i), \end{aligned} \tag{5.2}$$

where c was at position i and $|W| = i - r$. The index is retained. For $(\rho, i) \in \mathcal{B}_r(H)$, the inverse ordering is

$$(w_1, \dots, w_{r-1}, w_{r+1}, \dots, w_i, w_r, w_{i+1}, \dots, w_n). \tag{5.3}$$

Empty blocks are omitted, so the formula also applies when $i = r$.

The prefix set is unchanged. Under Θ , condition (4.2) becomes $u_j \mapsto w_j$ for every $1 \leq j \leq r$. Let $\mathcal{B}(T, \mathbf{e})$ denote the supporting states in this representation. Thus

$$|\mathcal{B}_r(H)| = M, \quad |\mathcal{B}(T, \mathbf{e})| = R(T, e). \tag{5.4}$$

The first identity also follows directly: there are $r!e(H)(n-r)!$ orderings with an edge first, and each has $n-r+1$ possible values of i .

5.3 Joining copies along a common edge

Lemma 5.2 (Joining along an edge). *Suppose $T = T_1 \cup T_2$, the edge e belongs to both T_1 and T_2 , and $V(T_1) \cap V(T_2) = e$. Then*

$$R(T_1, e) + R(T_2, e) \leq M + R(T, e). \tag{5.5}$$

The inequality holds for arbitrary r -graphs T_1, T_2 with the stated intersection; they need not be tight trees.

Proof. Fix the same ordering $\mathbf{e}$ in all three targets and work in $\mathcal{B}_r(H)$. We construct an injection

$$\mathcal{B}(T_1, \mathbf{e}) \setminus \mathcal{B}(T, \mathbf{e}) \longrightarrow \mathcal{B}_r(H) \setminus \mathcal{B}(T_2, \mathbf{e}). \quad (5.6)$$

Take a source state (ρ, i) , and let a be the least index in $\{r, \dots, i\}$ at which its prefix supports T_1 with the prescribed root-edge correspondence. Write

$$\rho = (E_0, A, B, C), \quad E_0 = (w_1, \dots, w_r), \quad A = (w_{r+1}, \dots, w_a), \quad B = (w_{a+1}, \dots, w_i).$$

Here E_0 lists the root-edge images, and A, B, C are consecutive lists of the remaining vertices. The original prefix ends after B . Set $b = r + |B|$ and define

$$((E_0, A, B, C), i) \longmapsto ((E_0, B, A, C), b). \quad (5.7)$$

In divider notation this is

$$(E_0, A, B \mid C) \longmapsto (E_0, B \mid A, C).$$

The image is a valid state of $\mathcal{B}_r(H)$, including when B is empty, since then $b = r$.

By the choice of a , the vertices in E_0, A contain a copy of T_1 with the specified images of the root vertices. If the image prefix E_0, B supported T_2 , the two embeddings would agree on every vertex of e and be disjoint outside e . Indeed, the non-root vertices of the first copy lie in A and those of the second lie in B . Their union would therefore embed T into the original prefix, contrary to the source condition. This proves that the image is in the right-hand side of (5.6).

To recover the source from an image, first read E_0 from the first r positions and B from the rest of the image prefix. Starting immediately after the image divider, find the shortest initial block A' such that E_0, A' supports T_1 with its prescribed root edge. Allow A' to be empty. The actual block A works. No shorter initial block works, since such a block would be a proper initial segment of A and would contradict the original minimality of a . Thus $A' = A$. This test uses E_0 and the block after the divider; it does *not* include B in the support test. The remaining entries are C , and the original index is $i = r + |A| + |B|$. Both the ordering and index have been recovered uniquely.

Finally, $\mathcal{B}(T, \mathbf{e}) \subseteq \mathcal{B}(T_1, \mathbf{e})$. Taking cardinalities in the injection and using (5.4) gives

$$R(T_1, e) - R(T, e) \leq M - R(T_2, e),$$

which is (5.5). □

Remark 5.3 (The case of an empty middle list). If B is empty, the image has $i = r$: its prefix consists just of the root edge. This value of i is included in $\mathcal{B}_r(H)$, so the map is still defined and its inverse still recovers A . No pairs need to be discarded in this case.

6 Completing the induction

We now show that every tight tree can be decomposed into the two cases treated above, whatever edge is chosen as the root. We do this by constructing an ordinary tree whose nodes are the hyperedges. This auxiliary tree records how the hyperedges are attached to one another.

6.1 A tree on the hyperedges

Lemma 6.1 (An auxiliary tree on the edges). *Let T be a tight r -tree. There is a tree $\mathcal{T}$ whose nodes are the hyperedges of T such that:*

- (i) adjacent nodes correspond to edges meeting in exactly $r - 1$ vertices;
- (ii) for each vertex x of T , the nodes whose edges contain x form a connected subtree of $\mathcal{T}$;
- (iii) after rooting $\mathcal{T}$ at any node, every parent-before-child ordering is a tight-tree construction order for T .

The edges indexed by any nonempty connected subtree of $\mathcal{T}$, with their union as vertex set, also form a tight r -tree.

Proof. Choose a tight-tree construction order $e_1, \dots, e_m$ and one permitted parent $e_{p(j)}$ for each $j \geq 2$. Join e_j to $e_{p(j)}$. Each new node attaches to a preceding node by one edge, so these connections form a tree. The child introduces exactly one new vertex, and all its other $r - 1$ vertices belong to the parent; this proves (i).

To prove (ii), follow the construction. A vertex first occurs in a single node. Whenever a subsequent edge contains that existing vertex, its parent must contain it too, since every child vertex other than the newly introduced one belongs to the parent. The nodes containing that vertex therefore remain connected at every step.

Now root $\mathcal{T}$ at any node and order its nodes with parents before children. Consider a child edge g with parent f , and let x be the unique vertex of $g \setminus f$. If x occurred in an earlier edge h , then h could not be a descendant of g , because descendants occur after g . The path from g to h would pass through f . Property (ii) would force $x \in f$, a contradiction. Thus x is new, and $g \setminus \{x\} \subseteq f$ proves (iii).

For a connected subtree, the nodes containing any given vertex are again connected whenever nonempty: the unique path between two such nodes lies in both the original vertex-subtree and the chosen subtree. The same parent-before-child argument proves the final assertion. $\square$

Lemma 6.2 (Decomposition at a distinguished edge). *Let T have $m \geq 2$ edges and distinguished edge e , and root $\mathcal{T}$ from Lemma 6.1 at e .*

- (i) *If e has one neighbour f in $\mathcal{T}$, the vertex of $e \setminus f$ occurs in no other edge of T . Removing e and that vertex leaves a tight tree S with $m - 1$ edges.*
- (ii) *If e has at least two neighbours, T can be written $T_1 \cup T_2$, with both T_j smaller tight trees containing e , such that*

$$V(T_1) \cap V(T_2) = e, \quad e(T_1) + e(T_2) = m + 1.$$

Proof. In (i), if the vertex $x \in e \setminus f$ occurred in another node, the path to that node would pass through f , contradicting property (ii) of Lemma 6.1. The remaining nodes form a connected subtree, so they give a tight tree by Lemma 6.1.

In (ii), partition the components of $\mathcal{T} - e$ into two nonempty groups and restore e to each group. Each group with e forms a connected subtree, and hence defines a tight tree T_j . If a vertex occurs in both groups, property (ii) of Lemma 6.1, applied to the path through e , forces that vertex to belong to e . Conversely, both trees contain all vertices of e . Their edge sets overlap in exactly the single edge e , which gives the displayed edge-count identity. Nonemptiness of both groups ensures that each T_j has fewer than m edges. $\square$

6.2 The counting inequality

Theorem 6.3 (The main counting inequality). *Let $r \geq 2$, let T be a tight r -tree with $m \geq 1$ edges, and distinguish any edge e of T . For every r -graph H on $n \geq r$ vertices,*

$$M \leq R_H(T, e) + (m - 1)Q, \quad (6.1)$$

where M and Q are given in (4.1).

Proof. Induct on m , simultaneously for all tight trees and all choices of their distinguished edges. If $m = 1$, every marked state supplies the prescribed copy of the sole edge, so $R_H(T, e) = M$.

Suppose $m \geq 2$ and use the decomposition of Lemma 6.2. If e has one neighbour f , write S for the remaining tight tree. The edge e is obtained from f by retaining its intersection with e and adjoining the vertex of e absent from S . Induction rooted at f , followed by Lemma 5.1, gives

$$M \leq R_H(S, f) + (m - 2)Q \leq R_H(T, e) + (m - 1)Q.$$

Otherwise write $T = T_1 \cup T_2$ as in part (ii) and put $m_j = e(T_j)$. Add the induction inequalities rooted at e and apply Lemma 5.2:

$$\begin{aligned} 2M &\leq R_H(T_1, e) + R_H(T_2, e) + (m_1 + m_2 - 2)Q \\ &\leq M + R_H(T, e) + (m - 1)Q. \end{aligned}$$

Subtracting M proves (6.1). Root-order choices made in the leaf step do not change the counts, by Lemma 4.2. $\square$

Proof of Theorem 1.2. Write $n = |V(H)|$. For $n < r$, the host has no edges and the inequality is immediate. Otherwise distinguish any edge e of T . The forbidden-copy hypothesis gives $R_H(T, e) = 0$. Substitute (4.1) into Theorem 6.3 to obtain

$$r! e(H)(n - r + 1)! \leq (m - 1)(r - 1)! |\partial H|(n - r + 1)!.$$

Division by the positive factorial factor $(r - 1)!(n - r + 1)!$ gives (1.1). This includes the one-edge target and the empty host: the argument does not divide by Q . $\square$

7 An example and sharpness

The following example illustrates why the copies must agree on every vertex of their common edge.

Example 7.1 (Branches through three different pairs). Let $r = 3$ and distinguish $e = 123$ in the tight tree

$$E(T) = \{123, 124, 135, 236\}.$$

Write $T_1 = \{123, 124\}$, $T_2 = \{123, 135\}$, and $T_3 = \{123, 236\}$, with the vertex sets understood to be their unions. For each j , start with the single edge other than 123 and apply Lemma 5.1, adding 123 as the new distinguished edge. The new vertices are respectively 3, 2, 1, so this gives

$$R_H(T_j, 123) \geq M - Q \quad (j = 1, 2, 3).$$

Two applications of Lemma 5.2 yield

$$R_H(T, 123) \geq R_H(T_1, 123) + R_H(T_2, 123) + R_H(T_3, 123) - 2M \geq M - 3Q.$$

The copies agree on the images of 1, 2, 3. The injection in Lemma 5.2 uses copies whose other vertices lie in disjoint parts of the permutation, so their union is a copy of T . The resulting bound for this particular four-edge tree also follows from [3].

Proposition 7.2 (Sharpness of the shadow coefficient). *For every $r \geq 2$, $m \geq 2$, and tight r -tree T with m edges, there is a T -free r -graph attaining equality in Theorem 1.2.*

Proof. Put $s = m + r - 2$ and take $H = K_s^{(r)}$. Since T has $m + r - 1 = s + 1$ vertices, it cannot embed in H . Moreover,

$$r \binom{s}{r} = (s - r + 1) \binom{s}{r-1} = (m - 1) \binom{s}{r-1}.$$

Here $|\partial H| = \binom{s}{r-1}$, giving the required equality. $\square$

Disjoint unions of these complete hypergraphs also attain equality in the shadow bound: a tight tree is connected, and the edge and shadow counts add over the components. Adding isolated host vertices leaves both counts unchanged.

When $r = 2$, the shadow consists of the singleton sets corresponding to nonisolated vertices. Theorem 1.2 therefore reads

$$2e(H) \leq (m - 1)|\{v \in V(H) : d_H(v) > 0\}|,$$

and implies the Erdős–Sós bound. As explained in the introduction, the method is adapted from [7]; the graph theorem itself is not used as a black box.

8 Lean formalization

The accompanying project in `lean/` proves the main theorem, its binomial consequence, and the counting inequality in Lean 4.19.0. It uses Lean’s bundled `Std` library and does not depend on `Mathlib`. We describe the definitions and their relation to the mathematical proof; instructions for checking the files are given in `lean/README.md`.

8.1 Definitions and theorem statements

The structure `FiniteHypergraph` represents a finite simple hypergraph by lists of vertices and edges. The vertex list and each edge list are duplicate-free, and no two stored edges are permutations of one another. Edge membership is defined up to permutation, so this representation imposes no order-preservation condition on embeddings. An `Embedding` is injective on the listed target vertices and maps each target edge to a host edge; nonedges need not be preserved.

The definition `IsTightTree` is the edge-order definition from Section 1: each edge after the first adds one new vertex, and its other vertices lie in an earlier edge. It also requires uniformity and excludes isolated target vertices. The decomposition and counting inequalities are proved, not assumed. The parameter m is the number of edges, and the identity $|V(T)| = m + r - 1$ is also proved in Lean.

The principal declarations are

- `Kalai.kalai_master`, corresponding to Theorem 6.3;
- `Kalai.kalai_shadow_bound`, corresponding to Theorem 1.2;
- `Kalai.kalai_bound`, corresponding to Corollary 1.3; and
- `Kalai.kalai_contains_tree`, giving the equivalent embedding conclusion above the binomial threshold.

The inequalities use natural numbers, with the denominator r cleared. The code counts each shadow set once, regardless of how many edges contain it. It defines the binomial coefficient by Pascal’s recurrence and proves the corresponding subset-counting identity. The statements include empty hosts and hosts with fewer than r vertices.

8.2 Relation to the proof

The formal proof establishes the two counting inequalities, the injection for joining copies and its inverse, and the formulas for M and Q . It also checks that reordering the chosen edge and moving the marking edge to the front leave the counts unchanged. As in the paper, pairs (π, i) are counted once, not once for each embedding they admit.

The decomposition argument differs from Section 6. Rather than construct the auxiliary tree on the hyperedges, Lean proves directly that every tight tree, with any chosen root edge, can be built from single edges by the leaf and joining operations. This is proved by induction from the original edge order, including the steps needed to insert an edge at any parent and to change the root. The result is then used to prove the main counting inequality. The separate proof for paths and the auxiliary-tree lemmas are not formalized line by line.

8.3 Verification

The project specifies the Lean version and includes the source files and a script that compiles them and checks which axioms the theorems depend on. The theorem statements are collected in `lean/Statements.lean`. The permitted logical axioms are `propext`, `Classical.choice`, and `Quot.sound`. Concrete formal tests cover root reorderings, branches through different faces, empty hosts, and a sharp graph-path example.

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